Glucose-Dependent Insulinotropic Polypeptide Is Associated With Lower Low-Density Lipoprotein But Unhealthy Fat Distribution, Independent of Insulin: The ADDITION-PRO Study

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Context: Glucose-dependent insulinotropic polypeptide (GIP) may increase lipid clearance by stimulating lipid uptake. However, given that GIP promotes release of insulin by the pancreas and insulin is anti-lipolytic, the effect may be indirect.

Objective: In this study we examined the association between GIP and lipid metabolism in individuals with low to high risk of type 2 diabetes and assessed whether the associations were modified by or mediated through insulin.

Design, Setting, and Participants: Analyses were based on the Danish cross-sectional ADDITION-PRO study (n = 1405). Lipid metabolism was measured by fasting plasma lipids and obesity including abdominal fat distribution assessed by ultrasonography. GIP and insulin were measured during an oral glucose tolerance test (0, 30 and 120 min). Linear regression analysis was used to study the associations between GIP, plasma lipids, and obesity measures.

Results: A doubling in fasting GIP levels was associated with lower low-density lipoprotein in both men (mean [95% CI] −0.10 mmol/l [−0.18–−0.03]) and women (−0.14 mmol/l [−0.23–−0.04]) and with higher high-density lipoprotein in women (0.06 mmol/l [−0.02–0.10]). In men, a doubling in stimulated GIP was associated with 0.13 cm less 0.01–0.25 sc fat but with more visceral abdominal fat (0.45 cm [0.12–0.78]) and higher waist-hip ratio (0.011 [0.004–0.019]).

Conclusions: Contrary to what was previously thought, GIP may be associated with improved low-density lipoprotein clearance but with an unhealthy fat distribution independent of insulin. The effect of GIP on obesity measures was substantially different between men and women. The potential effect of GIP on visceral and sc adipose tissue physiology warrants further examination. **(*J Clin Endocrinol Metab* 101: 485–493, 2016)**

Glucose-dependent insulinotropic polypeptide (GIP) is produced by the duodenal enteroendocrine K-cells and is released in response to upper intestinal nutrient exposure. The hormone potentiates glucose-induced insulin secretion and is partly responsible for the incretin effect (1). The release of GIP may be exaggerated in some

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Abbreviations: BMI, body mass index; CCK, cholecystokinin; CI, confidence interval; CT, computed tomography; FFA, free fatty acid; GIP, glucose-dependent insulinotropic polypeptide; GIPR, glucose-dependent insulinotropic polypeptide receptor; GLP, glucagon-like peptide; HbA_{1c}, glycosylated hemoglobin; HDL, high-density lipoprotein; LDL, low-density lipoprotein; LPL, lipoprotein lipase; MRI, magnetic resonance imaging; OGTT, oral glucose tolerance test; PPY, peptide YY; tAUC, total area under the curve; TG, triglyceride.

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type 2 diabetes populations after oral glucose ingestion (2–4) whereas it may be normal (5) or slightly reduced (6, 7) in others. Besides enhancing insulin secretion, GIP has been shown to stimulate growth and survival of the pancreatic β-cells and to have an effect on lipid metabolism (8). GIP regulates triglyceride (TG) turnover and may promote TG clearance from the blood by increasing deposition of fat in adipocytes (9, 10). Furthermore, GIP enhances the activity of lipoprotein lipase (LPL) and promotes insulin-dependent free fatty acid (FFA) uptake in adipocytes (11). GIP receptor (GIPR) knockout mice have reduced sc adipose tissue depots, improved insulin sensitivity, and seem to be resistant to diet-induced obesity (1, 12). However, the effect of GIP on human adipose tissue metabolism seems to differ depending on whether individuals are lean or overweight. GIP enhances FFA re-esterification in sc abdominal adipose tissue leading to increased fat storage and obesity in lean individuals during a hyperinsulinemic and hyperglycemic clamp (13). However, GIP does not induce any changes in TG uptake in obese individuals under similar experimental conditions, potentially due to a blunted adipose tissue blood flow (14). In patients with type 2 diabetes, higher fasting levels of GIP has been related to higher fasting and postprandial TG levels (2), indicating no substantial effect on plasma lipid clearance. In support of this finding, GIPR expression was demonstrated to be down-regulated in sc adipose tissue from obese patients, and GIPR signaling was disrupted in insulin-resistant states and in obesity (15). In vitro studies have shown that GIP may also stimulate lipolysis (8, 16) by which stored TG is hydrolyzed and released into the blood stream as FFA and glycerol. Furthermore, GIP has been shown to potentiate cytokine secretion (17), which, in combination with its prolipolytic response may impair insulin sensitivity and glucose uptake in human adipocytes (18). This dual adipogenic/lipolytic action of GIP seems counterintuitive and the clinical relevance remains elusive (Figure 1). McIntosh et al (8) suggested that GIP may act as a liposensor important for the maintenance of appropriate levels of circulating FFAs with and without the presence of insulin. During fasting, when insulin levels are low, a basal release of FFA stimulated by GIP may secure priming of β-cells for subsequent glucose stimulation (8). On the contrary, after exposure to nutrients, GIP stimulates release of insulin and both hormones may contribute to the stimulation of adipogenesis (8). As GIP promotes release of insulin by the pancreas and insulin is antilipolytic in adipocytes during insulin-sensitive states, some of the effects of GIP have been suggested to be indirect. However, GIP receptors are expressed in human adipose tissue (15) and are enriched during adipocyte differentiation (19), consistent with a direct effect of GIP on adipocyte metabolism.

To our knowledge, the association between GIP and lipid metabolism as well as the modifying role of insulin has not yet been examined in a large cohort of individuals with low to high risk of diabetes, for which reason we

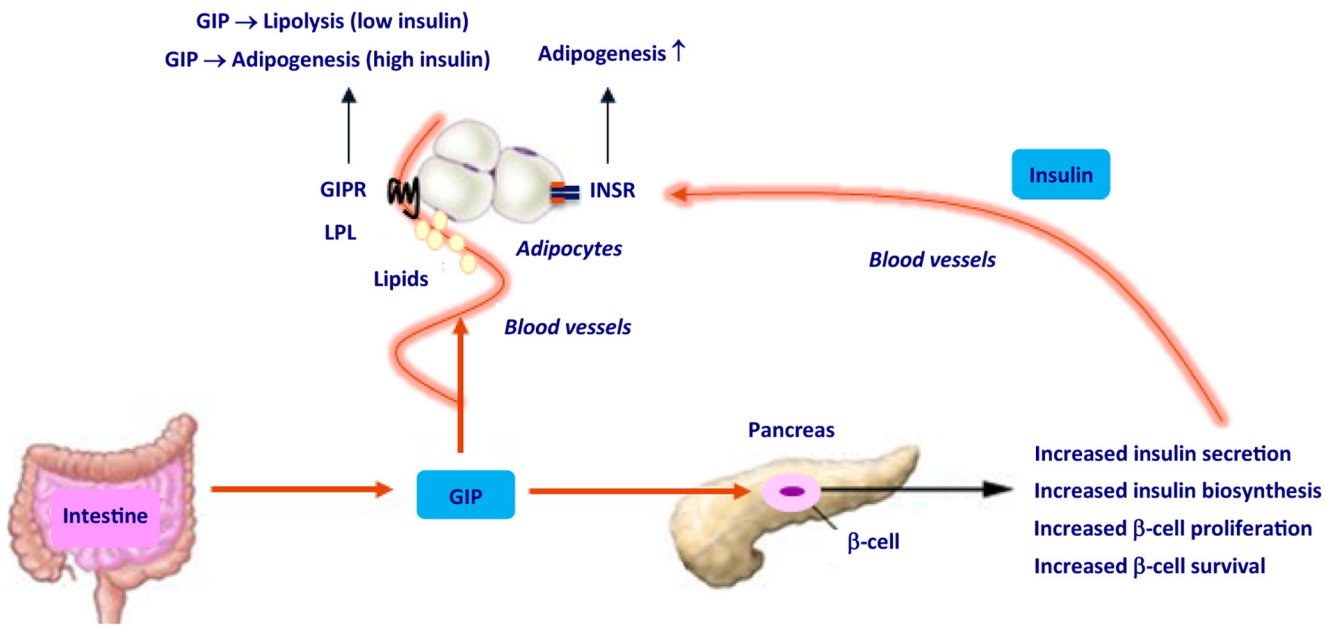

Figure 1. Actions of GIP on key tissues important for the control of glucose and lipid homeostasis. GIP promotes insulin biosynthesis, insulin secretion, and islet cell survival. Furthermore, GIP directly engages GIPRs on adipocytes and regulates lipid metabolism (potentially both lipolysis and adipogenesis depending on the level of insulin). The effect of GIP on adipose tissue is partially regulated through the activation of LPL. Increased insulin secreted upon GIP release inhibits adipocyte lipolysis and stimulates adipogenesis under insulin-sensitive states through the insulin receptor (INSR). Inspired by Drucker, 2007 (1).

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tested whether GIP has an insulin-independent effect on lipid metabolism in 1405 Danish participants.

Materials and Methods

Study population

Data obtained in this study was from the ADDITION-PRO cohort, a longitudinal risk-stratified group of individuals at low to high risk of developing type 2 diabetes. Participants were recruited using a population-based screening approach in Danish general practice (2001–2006) from which 16 136 suitable participants were selected (20). As previously described, screened individuals with impaired glucose regulation and a random subsample of individuals at lower diabetes risk were invited to a follow-up health examination (2009–2011), which 2082 participants (50% of invited) attended (20). The study was performed in accordance with the Helsinki Declaration and approved by the ethical committee of the Central Denmark Region (Journal No. 20080229). All participants provided written informed consent before participating in the study.

Clinical visit

After an overnight fast of at least 8 hours, blood samples for measurements of fasting serum insulin, plasma glycosylated hemoglobin (HbA_{1c}), TG, low-density lipoprotein (LDL), high-density lipoprotein (HDL), and total cholesterol were taken. Hereafter, a standard 75-g oral glucose tolerance test (OGTT) was given to all participants without known diabetes. Blood samples were drawn at 0, 30, and 120 minutes for assessment of serum insulin, plasma glucose, and GIP concentrations. Participants were classified as having normal glucose tolerance or impaired glucose regulation by the World Health Organization 2006 criteria. Physical activity was measured using a combined accelerometer and heart rate monitor (ActiHeart, CamNTech Ltd.). The full ADDITION-PRO study protocol is described elsewhere (20).

Biochemical measures

Blood samples for measurement of GIP were taken in tubes containing EDTA and put on ice immediately, centrifuged, and plasma was stored at −80 degrees. Total plasma GIP concentrations (the sum of intact GIP plus the metabolite GIP 3–42) were measured by RIA at the Department of Biomedical Sciences, University of Copenhagen, as previously described (21, 22). The analytical detection limit was 1 pmol/L and intra- and interassay coefficients of variation were 6.0 and 15%, respectively. The samples were analyzed consecutively during 2 months using identical quality controls and identical batches of all reagents. Serum insulin was measured using immunoassay (AutoDELFIA, PerkinElmer) and HbA_{1c} was determined by HPLC (TOSOH G7). Lipids were measured using the Vitros 5600 Integrated System (Ortho Clinical Diagnostics). Plasma for analysis of total cholesterol, HDL, and TG was collected in lithium-heparin-coated tubes, incubated for 0.5–1.5 hours at room temperature, and centrifuged at 3000 rpm for 10 minutes without cooling. LDL was calculated using the Friedewald's equation (23).

Body fat distribution

Body weight was measured without shoes to the nearest 0.1 kg with a Tanita Body Composition Analyzer (Tokyo, Japan), and height was measured to the nearest millimeter using a fixed stadiometer (Seca, Medical Scales and Measuring Systems). Waist circumference was assessed with a tape measure at the midpoint between the lower costal margin and the anterior superior iliac crest. Hip circumference was measured at the level of the greater trochanters. Abdominal fat distribution was assessed by ultrasonography (Logiq9 ultrasound machine, GE Healthcare) (24, 25). The transducer was placed on the abdomen and measurements were performed at the end of a quiet expiration using minimal pressure. Visceral fat thickness was measured as the depth (cm) from the peritoneum to the lumbar spine, and sc fat thickness as the depth (cm) from the skin to the linea alba as previously described (26, 27). Body fat percentage was assessed by bioelectrical impedance using the TANITA analyzer.

Calculations and statistical analysis

Statistical analyses were performed in R version 3.1.3 (The R Foundation for Statistical Computing) and SAS version 9.2 (SAS Institute). Participants with known diabetes (n = 336), those fasting less than 8 hours prior to the health examination (n = 20), those with no blood samples taken for measurement of GIP (n = 315), and those with no fasting insulin measurement (n = 6) were excluded, leaving 1405 individuals for analysis.

Using the trapezoid rule, stimulated GIP was calculated as total areas under the curves (tAUCs) from the basal state to 30 minutes. We used the total early area under the curve (0–30 min) as the main exposure variable in the analyses because this measure is considered a reflection of the metabolic effect of the hormone (2, 28). The incremental GIP response was not considered relevant in this context because it reflects the response of the stimulus. We used stimulated GIP to study the associations of GIP with obesity measures. However, given that we measure plasma lipids in the fasting state, we chose to associate lipids with the corresponding level of fasting GIP.

Insulin and GIP data were logarithmically transformed before analysis to fulfill the requirement of a normal distribution of the model residuals. The data are presented as means with SDs for normally distributed variables and as geometric means with corresponding 95% confidence intervals (CIs) for variables with a skewed distribution. All analyses were stratified by sex because it is well known that men and women display a different distribution of adipose tissue depots as well as in levels of plasma lipoproteins (29, 30).

Associations of fasting GIP with various measures of fasting lipid metabolism (plasma TG, LDL, HDL, and total cholesterol) and $tAUC_{0-30\ minutes}$ GIP with measures of obesity (waist circumference, waist/hip ratio, fat percentage, sc abdominal fat, visceral abdominal fat, and body mass index [BMI]) were studied using linear regression analyses. Because we hypothesized that the effect of GIP on lipid uptake is dependent on the insulin level of the individual, we adjusted for insulin but also tested for a modifying effect of insulin on the associations. Analyses were further adjusted for age, smoking, and physical activity. A two-sided level of significance of 5% was applied after adjustment for multiple testing using the Benjamini method (31).

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Results

Characteristics of the study population by sex and glucose tolerance status

Characteristics of the study population stratified by sex are shown in Table 1. Men in the ADDITION-PRO cohort were slightly older and more likely to be smokers than the female group Fasting levels of insulin, BMI, and waist circumference were higher in men compared with women. Women displayed a higher fat percentage, larger sc and smaller visceral adipose depots as well as higher levels of lipids (total cholesterol, LDL, and HDL) compared with men. Notably, women demonstrated a higher GIP release compared with men (30 min, 120 min). Characteristics of the study population by glucose tolerance status are shown in Supplemental Table 1 for women and Supplemental Table 2 for men. GIP levels did not differ by glucose tolerance status for any of the sexes (Figure 2).

Measures of obesity associated with GIP response during OGTT

In men, high levels of stimulated GIP ($tAUC_{0-30}$) were associated with higher waist-hip ratio (Figure 3) and higher levels of visceral abdominal fat (Figure 3). In contrast, the sc abdominal fat depot was lower in men with high levels of stimulated GIP ($tAUC_{0-30}$) (Figure 3). A doubling in GIP was associated with 0.13 cm less sc fat (95% CI, 0.01–0.25) but with 0.45 cm more (0.12–0.78) visceral abdominal fat and higher waist-hip ratio (0.011 [0.004–0.019]). In none of these associations did we find an interaction with insulin, indicating that the effect of GIP on male fat distribution is not modified by levels of insulin. In women, high levels of stimulated GIP were associated with a 1.2%-point lower (−2.0–−0.4) fat percentage (Figure 3). For any GIP level, obesity levels were in general highest for individuals with high insulin concentrations ($tAUC_{0-30}$ insulin) in both sexes. Analyses with BMI or waist circumference as outcomes showed similar results (Online Appendix and Supplemental Figure 1).

Associations of lipid measures with fasting plasma GIP response

High levels of fasting plasma GIP were associated with low plasma LDL cholesterol in both men and women (Figure 4). A doubling in plasma GIP levels was associated with 0.10 (0.03–0.18) mmol/L lower plasma LDL cholesterol in men and 0.14 (0.04–0.23) mmol/L lower plasma LDL cholesterol in women. Notably, the association was similar for all levels of insulin, showing no difference in participants with high vs low serum insulin concentrations. A doubling in plasma GIP levels was associated with a 0.06 mmol/L higher (0.02–0.10) plasma HDL cholesterol level in women, whereas GIP was not associated with HDL cholesterol levels in men. For a given fasting GIP level, the HDL cholesterol concentration was highest for women with lowest insulin concentrations (0.17 mmol/L difference between the lower and upper

Table 1. Characteristics of the Study Population by Sex

Characteristic	Women	Men	*P* Value
N	671	740	
Age, y	65.5 (7.5)	66.4 (6.8)	**.022**
Current smokers, %	14.0 (11.5–16.9)	18.4 (15.7–21.4)	**.026**
Serum insulin, pmol/L	35.4 (13.0–97.0)	38.8 (14.0–109.0)	**.007**
HbA_{1c}, %	5.7 (0.4)	5.7 (0.5)	.980
HbA_{1c}, mmol/mol	44.4 (4.4)	44.4 (5.4)	.980
Energy expenditure, $kJ \cdot d^{-1} \cdot kg^{-1}$	51.2 (21.2)	52.6 (26.1)	.281
BMI, kg/m^2	26.6 (5.1)	27.5 (3.8)	**< .001**
Waist circumference, cm	89.7 (12.9)	100.5 (10.9)	**< .001**
Hip circumference, cm	102.0 (11.3)	101.3 (7.8)	.166
Fat percentage	37.4 (6.8)	26.9 (6.1)	**< .001**
Subcutaneous abdominal depot, cm	2.8 (1.1)	2.3 (0.9)	**< .001**
Visceral abdominal depot, cm	6.8 (2.2)	8.9 (2.7)	**< .001**
TG, mmol/L	1.1 (0.6–2.3)	1.1 (0.6–2.5)	.109
Total cholesterol, mmol/L	5.6 (1.1)	5.3 (1.0)	**< .001**
LDL, mmol/L	3.4 (1.0)	3.3 (0.9)	**.026**
HDL, mmol/L	1.7 (0.4)	1.5 (0.4)	**< .001**
Plasma GIP, 0 min, pmol/L	8.6 (3.0–18.0)	8.2 (3.0–18.0)	.194
Plasma GIP, 30 min, pmol/L	51.3 (24.0–107.0)	45.3 (22.0–91.0)	**< .001**
Plasma GIP, 120 min, pmol/L	49.2 (24.0–92.0)	38.6 (18.0–77.0)	**< .001**
Plasma GIP, $tAUC_{0-30}$, pmol/L × min	915 (465–1800)	821 (420–1590)	**< .001**
Plasma GIP, $tAUC_{0-120}$, pmol/L × min	5527 (2925–10 230)	4696 (2490–8775)	**< .001**

Data are shown as means (SD), geometric means (95% CI), or percentages (95% CI).

P values in bold are significant after multiple testing adjustment.

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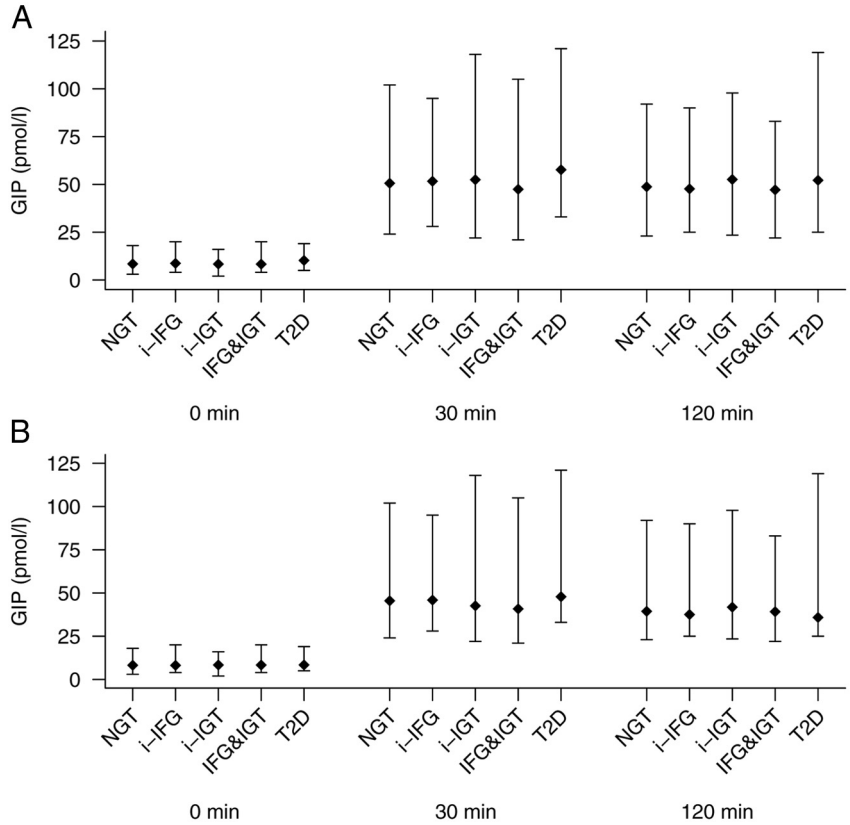

Figure 2. Levels of GIP by glucose tolerance status during OGTT for women (A) and men (B) separately. Data are geometric means with 95% CIs.

quartile of insulin). None of the above-mentioned associations showed interaction with insulin levels. However, we found a modifying effect of insulin on the association between fasting GIP and TG in women. When insulin was high (third quartile in the insulin distribution), a doubling in plasma GIP levels was associated with a 2.2% higher (−2.4–7.1) plasma TG. On the contrary, when insulin was low (first quartile in the insulin distribution), a doubling in plasma GIP levels was associated with an 8.5% lower (−12.5–−4.4) plasma TG.

Discussion

Using the ADDITION-PRO cohort, our aim was to examine the association of GIP with lipid metabolism in individuals with low to high risk for developing type 2 diabetes. Results obtained in this study demonstrate that high fasting plasma GIP levels are associated with low fasting levels of plasma LDL cholesterol in both men and women and with increased levels of HDL cholesterol in women. The effect of GIP on obesity measures was substantially different between men and women. In women, high GIP was only significantly associated with a lower fat percentage. In contrast, in men, high levels of GIP were associated with more visceral fat and a higher waist-hip ratio as well as less sc fat. After adjusting for insulin levels the associations mentioned above were still statistically significant, indicating that GIP has an effect on fat distribution independent of insulin. With the exception of TG, there was no interaction with insulin, which shows that the associations between GIP and fat metabolism were of similar magnitude across the entire range of fasting insulin secretion.

GIP and plasma lipids

White adipose tissue has the largest energy storage-capacity of the human body. In white adipose tissue lipids are stored as TG in lipid droplets of the adipocyte (32). The uptake and release of lipids in adipose tissue is a highly regulated process, which plays an essential role in energy homeostasis. Lipoproteins of TG, cholesterol, and protein occur in blood with densities varying from very low-density lipoproteins (chylomicrons and very low-density lipoprotein) to high-density lipoproteins (HDL). TG in the circulating lipoproteins is hydrolyzed by LPL allowing fatty acids to diffuse into the adipocytes from the blood stream (32). GIP has been reported to regulate TG turnover and promote lipid clearance from the blood (9, 10) by enhancing the activity of LPL (11). The direct effect of GIP and the GIPR in adipose tissue has previously been demonstrated in lean individuals, but a similar effect could not be demonstrated in obesity and type 2 diabetes (2, 13–15). Using the ADDITION-PRO cohort we now show that fasting GIP is associated with lower LDL levels, an association which, at least to some extent, seems independent of fasting insulin levels.

GIP and the distribution of adipose tissue

Although GIP may induce lower levels of LDL, the collective effects of GIP on body fat distribution seem metabolically inexpedient, particularly in men. It is well known that individuals with excessive abdominal visceral fat have increased risk of developing type 2 diabetes (32). Likewise, a recent large case-cohort study has shown that waist circumference is a strong and BMI-independent predictor of type 2 diabetes incidence (27, 33). If GIP specifically increases FFA uptake in visceral adipose tissue in men, it may worsen the insulin resistant state of prediabetic individuals or patients with type 2 diabetes. Both visceral and sc

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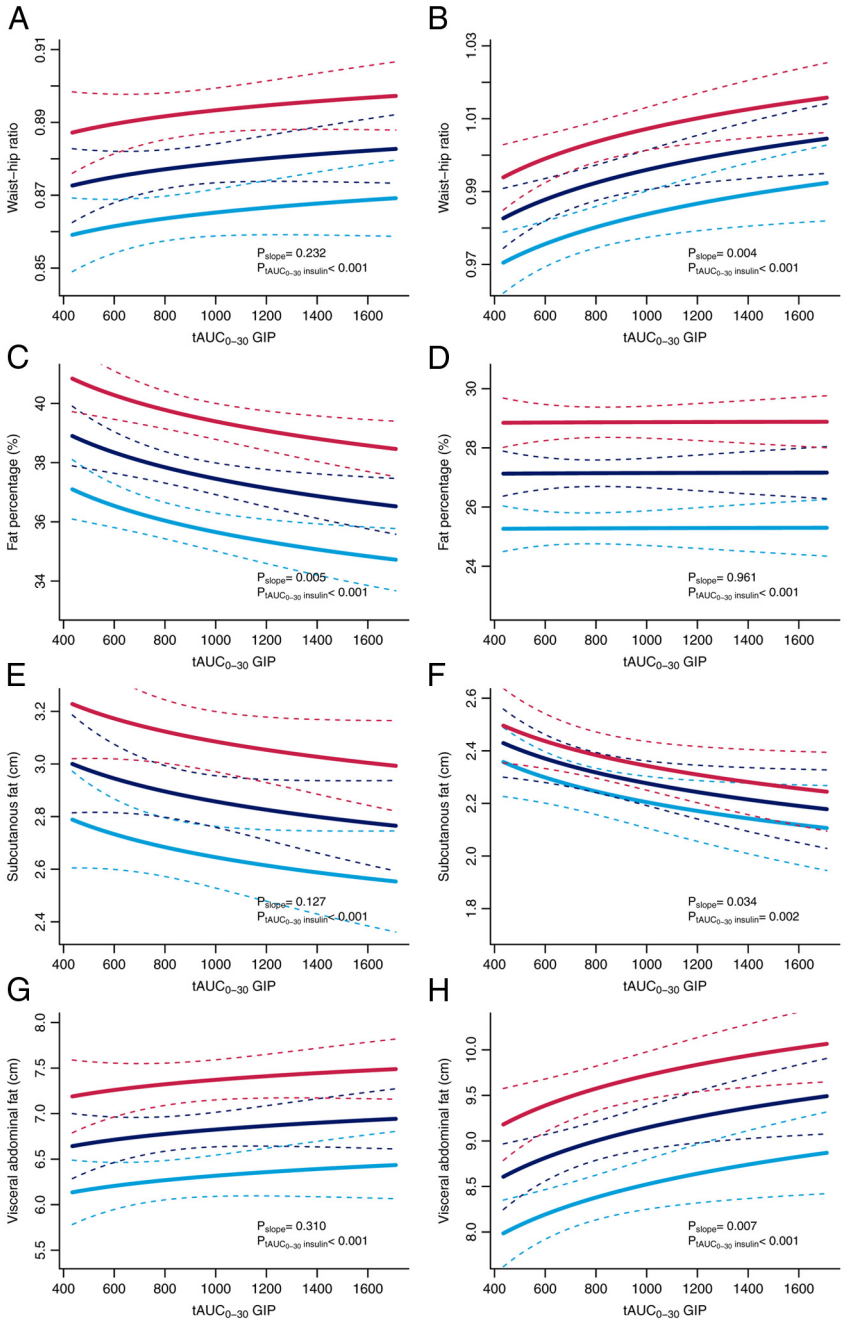

Figure 3. Associations of measures of obesity with $tAUC_{0-30}$ for GIP response during OGTT for women (A, C, E, G) and men (B, D, F, H) separately. Curves are shown for a hypothetical nonsmoking participant, age 66 years and physical activity energy expenditure of 52 ($kJ \cdot d^{-1} \cdot kg^{-1}$) (population means) at three different levels of $tAUC_{0-30}$ for serum insulin: the median (dark blue), lower quartile (light blue), and upper quartile (red). Dashed lines are 95% CIs. P_{slope}, significance level for test of slopes being different from 0. $P_{tAUC\ 0-30\ insulin}$, significance level for test of level of outcome being different for different levels of serum insulin ($tAUC_{0-30}$).

adipose tissues have great storage capacities for lipids (34). However, in lean men and women, a higher FFA uptake has been shown in visceral vs sc adipose tissue (34, 35). Moreover, it has been documented that visceral fat is more responsive to lipolytic stimuli such as catecholamines compared with sc fat (36, 37). This implies that fat stored in the visceral adipose tissue may be rapidly released again to the blood stream when needed. Importantly, the reduced GIP release observed in some patients with type 2 diabetes (6, 7) may be a protective mechanism in terms of reducing visceral adipose tissue storage and improving insulin sensitivity. In our study, however, we did not find lower levels of GIP in screen-detected type 2 diabetes.

It should be noticed that there are limitations when measuring visceral adipose tissue and sc adipose tissue by ultrasonography. It can be argued that 2-D or 3-D imaging techniques such as magnetic resonance imaging (MRI) or computed tomography (CT) are more suitable than ultrasound to capture the actual volume of abdominal visceral vs sc fat. The use of MRI or CT however, was not possible as these techniques are both costly and difficult to apply to large-scale studies such as the ADDITION-PRO. Ultrasonography offers an accessible and simple method for one-dimensional measure of abdominal fat distribution, and has been validated against MRI and CT. Additional functional studies are needed to examine the differential effect of GIP on visceral and sc adipose tissue physiology.

Dual effect of GIP on TG depending on circulating insulin

Another aim of this study was to uncover whether GIP has dual effects in terms of lipid uptake and lipolysis depending on the circulating insulin levels (Figure 1) (8). If so, we would expect to see a difference in TG levels depending on the levels of insulin. This was not the case for men. However, for women we observed a modifying effect of insulin regarding the association between GIP and TG. When insulin levels were high, high GIP was associated with increased levels of TG in the blood stream. On the contrary, when insulin levels were low, high GIP was associated with decreased levels of TG in the blood stream. This association is the opposite of what was hypothesized by McIntosh et al (8). McIntosh suggested that during low insulin (fasting), a basal release

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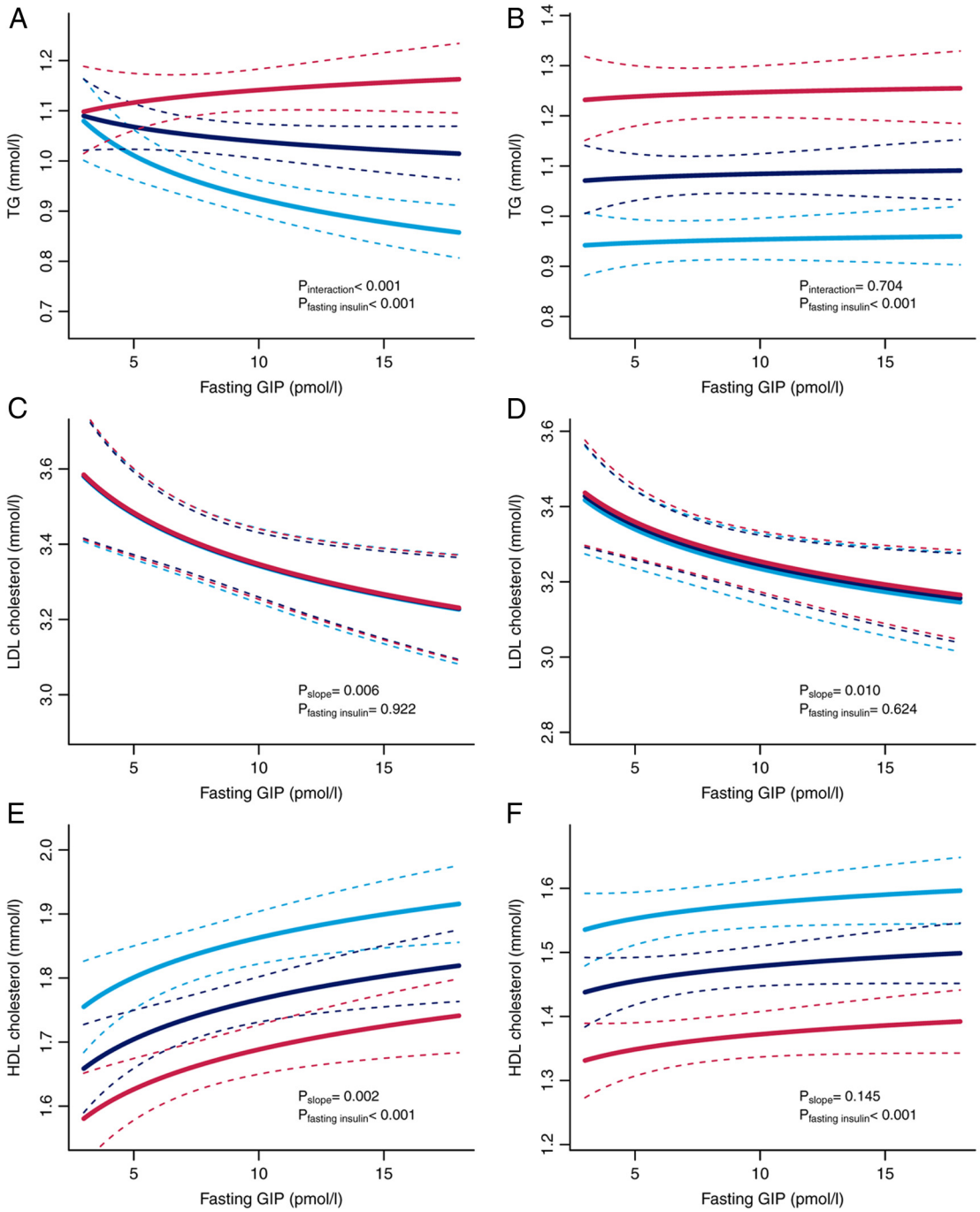

Figure 4. Associations of lipid measures with fasting GIP response during OGTT for women (A, C, E) and men (B, D, F) separately. Curves are shown for a hypothetical nonsmoking participant, age 66 years and physical activity energy expenditure of 52 ($kJ \cdot d^{-1} \cdot kg^{-1}$) (population means) at three different levels of fasting serum insulin: the median (dark blue), lower quartile (light blue), and upper quartile (red). Dashed lines are 95% CIs. P_{slope}, significance level for test of slopes being different from 0. $P_{fasting\ insulin}$, significance level for test of level of outcome being different for different levels of serum insulin.

of FFA stimulated by GIP may secure priming of β-cells for subsequent glucose stimulation (8). One may speculate that increased release (or reduced uptake) of TG and FFA induced by high GIP when circulating insulin concentrations are high may also prime β-cells for further insulin release. This effect may be beneficial in the metabolic healthy state. However, during prediabetes or type 2 diabetes this effect may increase lipotoxicity and insulin resistance.

Strengths and limitations

ADDITION-PRO entails the largest study population to date, which has been analyzed for GIP release and associations with circulating lipids as well as obesity (total of

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1405 individuals). ADDITION-PRO has the main strength that it contains simultaneous measures of insulin, GIP, and parameters of lipid metabolism in a large group of individuals with normal glucose tolerance, prediabetes, and screen-detected diabetes. However, other gut hormones not measured in this study (eg, glucagon-like polypeptide 2, cholecystokinin (CCK), peptide YY (PYY)), are also relevant for the regulation of energy balance. It is known that both CCK and PYY influence central regulation of energy balance and therefore may interfere with our interpretations (38). Likewise, it would have been relevant to include leptin in our analysis as leptin reflects fat storage in the adipose tissue as well as contributes to the regulation of appetite and energy expenditure centrally. Due to the cross-sectional nature of the study design, causal inference cannot be made. As such we cannot conclude on the causality between obesity and lipid metabolism and GIP. Future studies are needed to investigate this.

GIP's therapeutic potential

It has been debated whether GIP has a therapeutic potential for the treatment of human diabetes. In type 2 diabetes patients, the β-cell response to GIP is decreased and GIP (in contrast with glucagon-like peptide 1 [GLP-1]) has no insulinotropic activity (39). Furthermore, GIP has been shown to increase levels of glucagon in addition to insulin and to reduce levels of GLP-1, which collectively leads to increased levels of postprandial blood glucose (40). Based on results obtained in this present study, GIP may improve lipid clearance (lowering LDL levels), which hypothetically could reduce lipotoxicity and improve whole-body insulin sensitivity in both men and women. However, GIP is also associated with an unhealthy distribution of fat depots in men: reduced sc adipose tissue and larger visceral adipose tissue depots. Taking the above into account, GIP does not seem to be a strong therapeutic candidate for treatment of obesity-associated dysmetabolism.

GIP and the sexes

The differential associations of GIP with lipid metabolism seen between men and women are intriguing. One may speculate that GIP could play a role in the sex differences in the distribution of sc and visceral adipose tissue depots. Furthermore, GIP levels or GIP metabolism may partly explain why some people seem to preferentially store fat subcutaneously and others viscerally. Notably, we find an overall difference in GIP release between men and women.

Conclusion

In both men and women, GIP is associated with lower LDL levels and in men also with altered fat distribution independent of insulin. The potential effect of GIP on visceral and sc adipose tissue physiology warrants additional examination.

Acknowledgments

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Author Contributions: C.L.M. drafted the manuscript; D.V. performed the statistical analysis and C.L.M., D.V., K.F., and N.B.J. developed the analysis plan. The ADDITION-PRO study was designed by N.B.J., D.R.W., and T.L. C.L.M., D.V., K.F., N.B.J., D.R.W., A.J., O.P., T.H., T.L., M.E.J., S.S.T., and J.J.H. contributed to the interpretation of data, revised the manuscript critically, and approved the final version. J.J.H. and S.S.T. provided the GIP data. D.V. is guarantor of the work, had full access to all data in the study, and takes responsibility for the integrity of the data and the accuracy of the data analysis.

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